CSE345 · INFORMATION SYSTEM DESIGN & SOFTWARE ENGINEERING

System Analysis and Design of Daraz Bangladesh

Studying a real e-commerce platform to find its biggest problems — and designing a better system to fix them.

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Full report online: daraz-sad.vercel.app

WHAT WE'LL COVER TODAY

Presentation Roadmap — 4 Chapters, 4 Voices

1

Chapter 1

Recognition of Need

What Daraz Bangladesh is, and the 5 problems we found

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2

Chapter 2

Initial Feasibility Study

Is each problem realistic and worth fixing?

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Chapter 3

Analysis

How we gathered proof, and what the data shows

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Chapter 4

Design

Our proposed system, database, and new forms

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Daraz Bangladesh, and Why It's Worth Studying

Daraz Bangladesh is an online marketplace where people buy almost anything — electronics, clothes, groceries, and more. It connects thousands of independent sellers with millions of buyers, and it is owned by the Alibaba Group.

Because Daraz is large, well known, and well documented in reviews and news, it is an ideal real-life case for a System Analysis and Design study.

2015
Started operating in Bangladesh

#1
Largest online marketplace in BD

1000s
Independent sellers on the platform

CHAPTER 1 · RECOGNITION OF NEED

Five Problems We Found in Customer Reviews

1

Product Authenticity

Fake, damaged, or wrong products from unchecked sellers

2

Refund & Return Delays

Slow, unclear refunds with no tracking or reason given

3

Account Security & Fraud

Hacked accounts and fake Daraz look-alike pages

4

Delivery Service

Late orders and disputed "failed delivery" statuses

5

Customer Support

Hard to reach a real agent; slow seller replies

Five Simple Checks Before Fixing Anything

Before designing a fix, we asked: is it realistic? We checked every problem against five standard feasibility areas.

1

Technical

Does the technology to fix it already exist?

2

Economic

Do the benefits outweigh the cost?

3

Operational

Will it fit how Daraz already works?

4

Legal

Does it follow data & consumer laws?

5

Schedule

Can it be built in a reasonable time?

Result: All Five Problems Are Worth Fixing

EASIER & FASTER

These build on tools Daraz already has:

- **Refund & Return Process**
- **Delivery Service**
- **Customer Support**

MORE DEMANDING

These need new tools and take more effort:

- **Product Authenticity**
- **Account Security & Fraud**

Still realistic — just needs a bigger investment, like fraud detection and seller verification tools.

How We Gathered Our Information

1 Literature & Procedure Review

Studied Daraz's app, forms, policies, and industry best-practices from Amazon, Alibaba, and local rivals.

2 On-Site Observation

Used the Daraz app and website ourselves — from browsing to checkout to tracking an order.

3 Interviews

Called Daraz's real customer care helpline, plus talked to a seller, a customer, and a delivery rider.

4 Questionnaires

Built customer, seller, and app-user questionnaires, then matched them against public reviews.

What the Data Told Us

Top complaint categories in reviews



Delivery and refund complaints dominate — that's where the fix must focus first.

DFD of the Existing System

An 8-process data flow diagram — from Browse & Search to Deliver Order and Return & Refund — maps exactly where each of the 5 problems occurs in the workflow.

Cost-Benefit & Break-Even

≈ 11,429

order interactions to break even on the $40,000 investment — realistic within a short time for a platform this size.

CHAPTER 4 · DESIGN

Two Candidate Systems — We Picked the Stronger One

CANDIDATE 1 · SELECTED

Comprehensive Automated Trust & Security System

- Full seller ID & document verification
- Live Trust Score for every seller
- Fraud risk-scoring engine + OTP
- Guaranteed support escalation deadline

CANDIDATE 2

Incremental, Lower-Cost System

- Simple trust tiers, no new documents
- Rule-based OTP checks only
- Delivery OTP without photo proof
- Escalation without a strict deadline

A weighted scoring matrix gave Candidate 1 the higher score — it protects customers exactly where our analysis showed the biggest risks.

Inside the Proposed System

New DFD Processes

- Seller Verification & Trust Score
- Fraud Detection + OTP at checkout
- Delivery Proof Capture (code/photo)
- Digital Return Receipt
- Support Ticket Escalation (SLA)

Database Design

- Customer, Seller, Product
- Order, OrderItem, Payment
- Delivery (proof fields)
- ReturnRefund (receipt field)
- SupportTicket (SLA field)

New Forms

- Seller Verification form
- OTP / Account Security form
- Return & Refund Request form
- Rider App: Delivery Proof form

SEE IT LIVE

The Full Report, Presented Online

Every chapter, chart, diagram, and form from this project is laid out on our project website:

daraz-sad.vercel.app

Feel free to explore it after the presentation for the complete charts, DFDs, ER diagram, and wireframes.

WRAPPING UP

The Big Picture

Daraz's business model is sound. The gap is in “exception handling” — the tools that kick in once something goes wrong: a fake product, a hacked account, a disputed delivery, a stuck refund.

1

Stronger Trust

Verified sellers & visible trust scores

2

Safer Accounts

Fraud detection and instant OTP checks

3

Clearer Delivery

Proof of delivery ends false disputes

4

Faster Refunds

Trackable status with digital receipts

THANK YOU

Questions?

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